National University of Computer and Emerging Sciences [redacted]

Lahore Campus

Technical & Business Writing (SS2007/ SS2012)

Date: September 20th 2025

Course Instructor(s)

Mr. Muhammad Zulfiqar Ali

Sessional-I Exam

Total Time (Hrs): 1

Total Marks: 40

Total Questions: 2

Do not write below this line

Attempt all the questions.

Roll No	7ABC, 9A Sections	Student Signature

CLO #1: Use accurate paraphrasing, citations, and mechanics in technical writing contexts.

Q1: A. The following is a passage with generic content. Assume any realistic and relevant (quantitative) data to **construct** it into a professional, good quality, technical piece of writing. Do not change the original meaning, but you may add up any justifiable information. [10 marks]

'The purchased equipment for our grid station, which was just landed by the deliverers of our trusted suppliers, was very carefully packed in big cartons. There were many lying around, of various sizes and weights. I signed up to receive all the items as I was the in-charge officer. All my staff had gone as it was a holiday tomorrow, and the new shift will come the next day.'

B. Reorganize and paraphrase the given passage. The references are provided on the table. **Incorporate** these citations per IEEE format in your paraphrase where most appropriate. [10 marks]

'A Microgrid can be deployed to sustain electricity service when the hosting bulk power grid (to which the microgrid is connected) becomes unavailable.[1] Technologies validated through simulations and field demonstrations have shown that a microgrid can operate smoothly in a utility-connected or an islanded mode and during its transition between modes.[2] Electric energy in a microgrid is often produced by Distributed Energy Resources (DERs), including renewable energy and energy storage facilities as well as traditional synchronous generators of ratings up to a few MWs.[4] As the level of penetration of renewable energy becomes higher, the need for improving primary frequency response using inverter-based resources (IBRs) becomes essential to support power grid operation.'[3]

[1] M. Rasheduzzaman, J. A. Mueller, and J. W. Kimball, "Reduced-order small-signal model of microgrid systems," *IEEE Trans. Sustain. Energy*, vol. 6, no. 4, pp. 1292–1305, Oct. 2015.
[2] J. Eto et al., "The CERTS microgrid concept, as demonstrated at the CERTS/AEP Microgrid test bed," Lawrence Berkeley Nat. Lab., Berkeley, CA, USA, Tech. Rep., 2018.
[3] K. P. Schneider et al., "Improving primary frequency response to support networked microgrid operations," *IEEE Trans. Power Syst.*, vol. 34, no. 1, pp. 659–667, Jan. 2019.
[4] R. H. Lasseter, Z. Chen, and D. Pattabiraman, "Grid-forming inverters: A critical asset for the power grid," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 8, no. 2, pp. 925–935, Jun. 2020.

CLO # 2: Write effective technical content following ethical principles.

Q2 i. Assume that you work at a manufacturing company. During your routine inspection of the plant, you found some fault in one of the newly installed components of a machinery *(assume any realistic scenario)*. **Write a memo** to your senior position briefing him/her about the problem, steps to rectify it, and the expected performance after fixing it. Be precise, objective, and clear in writing your technical descriptions. Assume and mention any relevant data to support your point. [15 marks]

ii. Ethically, would you **express** your concerns and report to your seniors about the faulty and sub-standard quality of the component you mentioned in the above-written memo. Why or why not? Justify your answer with two strong reasons. [05 marks]